

SuperKart Project

Post Graduate Program in Artificial Intelligence and Machine Learning

Prepared by Christian Caceres

June 6, 2026

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Executive Summary

An exploratory data analysis (EDA) was conducted to identify the key factors influencing SuperKart's sales performance across products, stores, and regions. The analysis revealed that product attributes, store characteristics, and location-based factors significantly impact sales outcomes. Variables such as product visibility, weight, MRP, outlet type, outlet size, and outlet location tier emerged as important drivers of revenue, while notable sales variations across store formats highlighted opportunities for more targeted inventory and merchandising strategies.

To address these business challenges, multiple machine learning regression models were developed and evaluated to forecast product-store sales revenue. Model performance was assessed using industry-standard metrics, including RMSE, MAE, and R^2 , to ensure both predictive accuracy and generalization on unseen data. Among the models tested, the Tuned XGBoost Regressor delivered the strongest overall performance, achieving the lowest prediction error and highest predictive accuracy. Based on its superior results, the Tuned XGBoost Regressor was selected as the final model for deployment. The deployed solution provides SuperKart with a scalable, data-driven forecasting capability that supports inventory optimization, demand planning, and more informed business decision-making across its retail network.

Business Problem Overview and Solution Approach

Problem Overview:

SuperKart operates across multiple stores, regions, and product categories, making accurate sales forecasting critical for inventory management and business planning. Variations in demand, products, pricing, and store characteristics make manual forecasting unreliable and difficult to scale:

Key Challenges:

- Demand variability across stores and products.
- Inventory inefficiencies, including stockouts and overstocking.
- Forecasting methods that do not scale across regions.

Solution Approach:

A machine learning-based sales forecasting solution was developed to predict product-store sales revenue using historical data, product attributes, and store characteristics. The project included data preparation, feature engineering, and Exploratory Data Analysis (EDA) to identify key sales drivers and patterns.

Multiple regression models were built, evaluated, and optimized to determine the best-performing forecasting model. The selected model was then deployed through a backend API and user-friendly frontend application, enabling business users to generate accurate sales forecasts and support inventory, supply chain, and operational planning across SuperKart stores.

Data Overview

```
data.info()
```

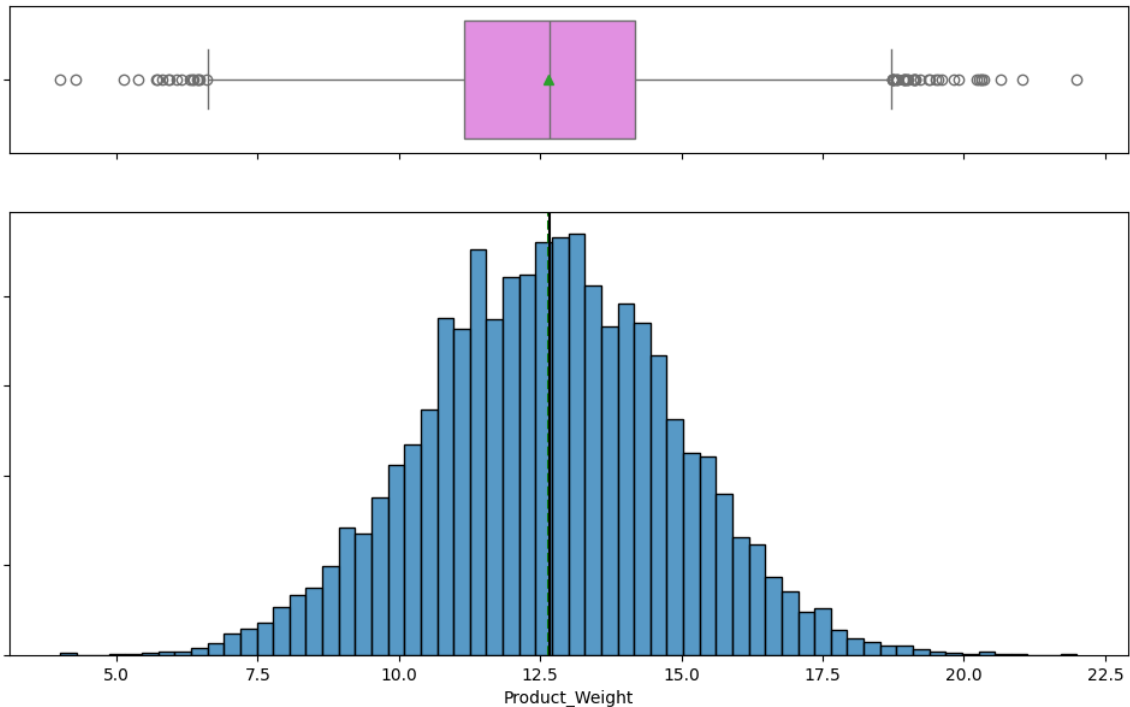
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8763 entries, 0 to 8762
Data columns (total 12 columns):
#   Column                      Non-Null Count  Dtype
---  -
0   Product_Id                  8763 non-null   object
1   Product_Weight              8763 non-null   float64
2   Product_Sugar_Content       8763 non-null   object
3   Product_Allocated_Area      8763 non-null   float64
4   Product_Type                8763 non-null   object
5   Product_MRP                 8763 non-null   float64
6   Store_Id                    8763 non-null   object
7   Store_Establishment_Year    8763 non-null   int64
8   Store_Size                  8763 non-null   object
9   Store_Location_City_Type    8763 non-null   object
10  Store_Type                  8763 non-null   object
11  Product_Store_Sales_Total   8763 non-null   float64
dtypes: float64(4), int64(1), object(7)
memory usage: 821.7+ KB
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Product_Id	8763	8763	FD306	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Weight	8763.0	NaN	NaN	NaN	12.653792	2.21732	4.0	11.15	12.66	14.18	22.0
Product_Sugar_Content	8763	4	Low Sugar	4885	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Allocated_Area	8763.0	NaN	NaN	NaN	0.068786	0.048204	0.004	0.031	0.056	0.096	0.298
Product_Type	8763	16	Fruits and Vegetables	1249	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_MRP	8763.0	NaN	NaN	NaN	147.032539	30.69411	31.0	126.16	146.74	167.585	266.0
Store_Id	8763	4	OUT004	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Establishment_Year	8763.0	NaN	NaN	NaN	2002.032751	8.388381	1987.0	1998.0	2009.0	2009.0	2009.0
Store_Size	8763	3	Medium	6025	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Location_City_Type	8763	3	Tier 2	6262	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Store_Type	8763	4	Supermarket Type2	4676	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Store_Sales_Total	8763.0	NaN	NaN	NaN	3464.00364	1065.630494	33.0	2761.715	3452.34	4145.165	8000.0

- The dataset contains 8,763 rows and 12 columns
- Target Variable: Product_Store_Sales_Total
- Missing Values: None
- Duplicated Values: None
- Categorical variables require encoding before modeling
- **Product Insights**: Low Sugar products dominate the inventory, Fruits and Vegetables are the largest product category, and Product_MRP shows a wide price range, making it a likely key driver of sales.
- **Store Insights**: The data is concentrated in Medium-sized stores, Tier 2 cities, and Supermarket Type2 locations, indicating potential imbalances that may influence sales patterns and model performance.
- **Sales Performance: Product_Store_Sales_Total** exhibits substantial variability (33–8,000 units sold), suggesting strong opportunities for machine learning models to identify the factors that drive high and low sales performance.

EDA Results

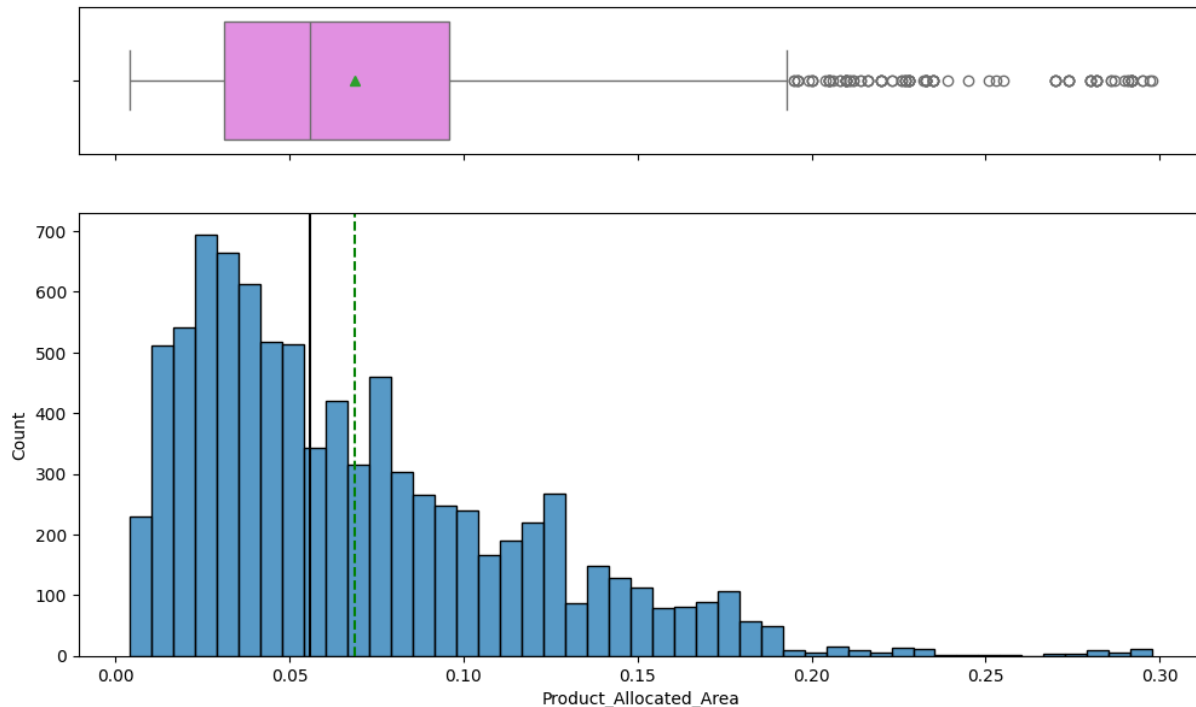
Univariate Analysis: Product Weight



- Product weight is approximately normally distributed, centered around 12.5–13.
- Mean and median are very close, indicating minimal skewness in the main distribution.
- Several outliers exist on both the lower and higher ends of the weight range.
- Most products fall within a relatively narrow weight range, suggesting standardized packaging sizes

EDA Results

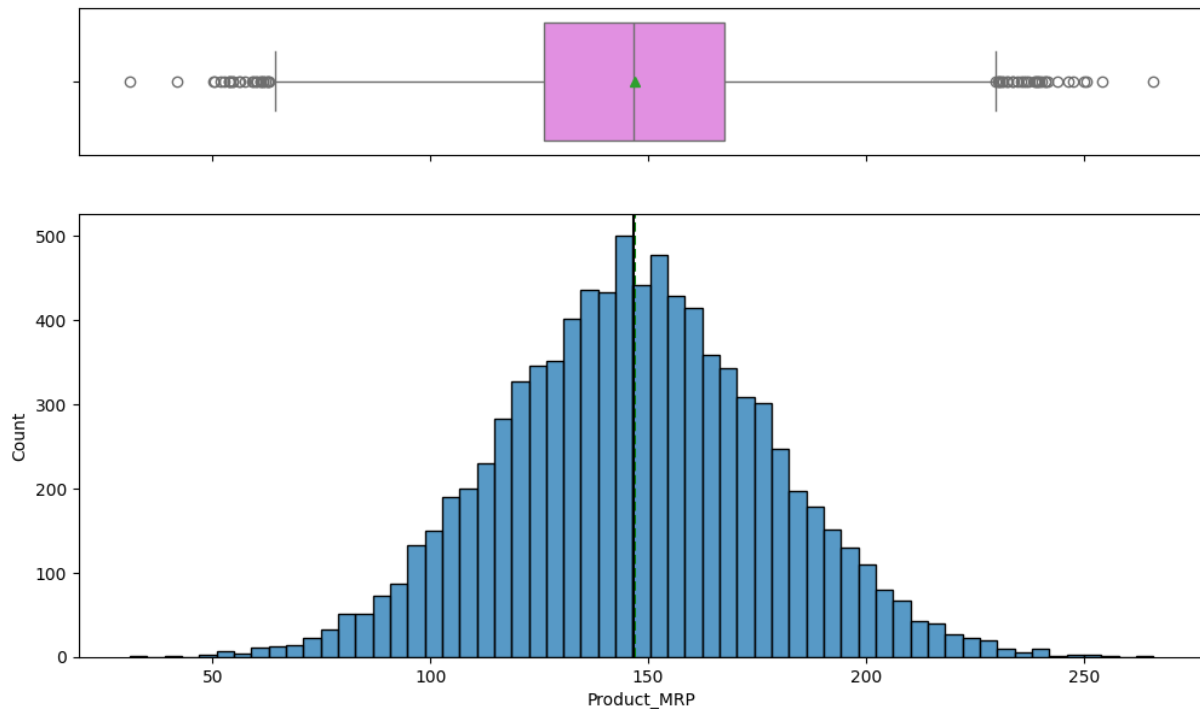
Univariate Analysis: Product Allocated Area



- Product allocated area is positively skewed, with most products receiving smaller shelf space allocations.
- Mean is higher than the median, indicating the presence of high-value outliers with significantly larger allocated areas.
- Several extreme outliers exist above 0.20, suggesting a small number of products dominate shelf space.
- Majority of products are concentrated below 0.10 allocated area, indicating limited shelf distribution for most items.

EDA Results

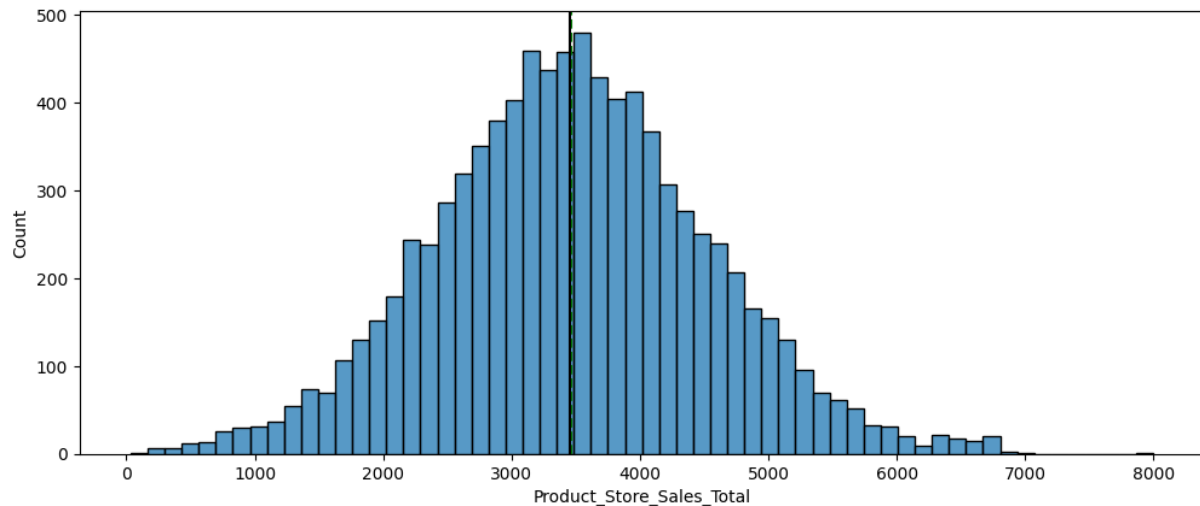
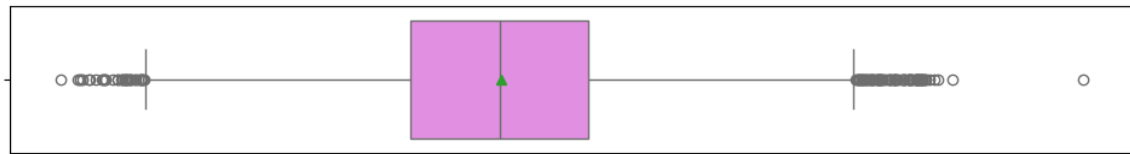
Univariate Analysis: Product MRP



- Product MRP is approximately normally distributed, centered around 145–150. Mean and median are closely aligned, indicating a fairly symmetric distribution.
- Most products fall within the mid-price range, suggesting balanced pricing across the catalog.
- A few low and high price outliers exist, indicating the presence of budget and premium product segments.

EDA Results

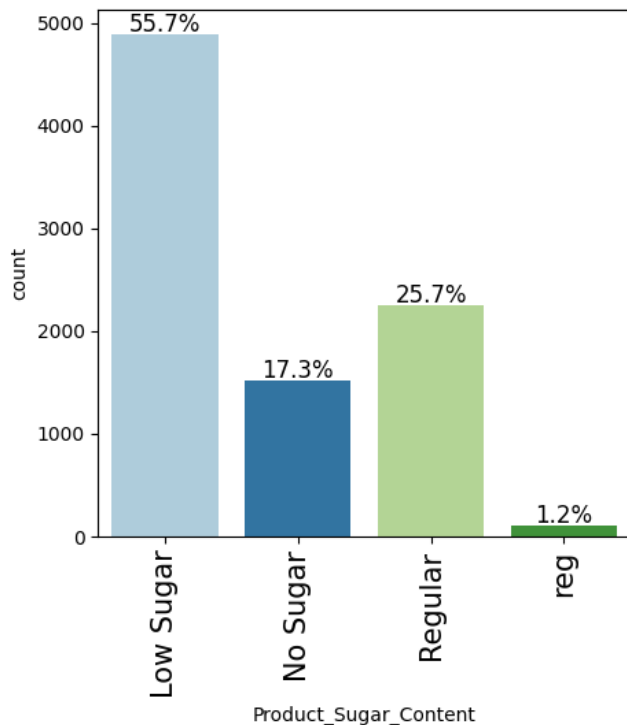
Univariate Analysis: Product Sales Total



- Product store sales are approximately normally distributed, centered around 3,400–3,600.
- Mean and median are closely aligned, indicating a relatively symmetric distribution.
- Most products generate sales within a moderate range, with fewer extremely low or high sales values.
- Several outliers exist on both ends, indicating a small number of products significantly underperform or outperform the average.

EDA Results

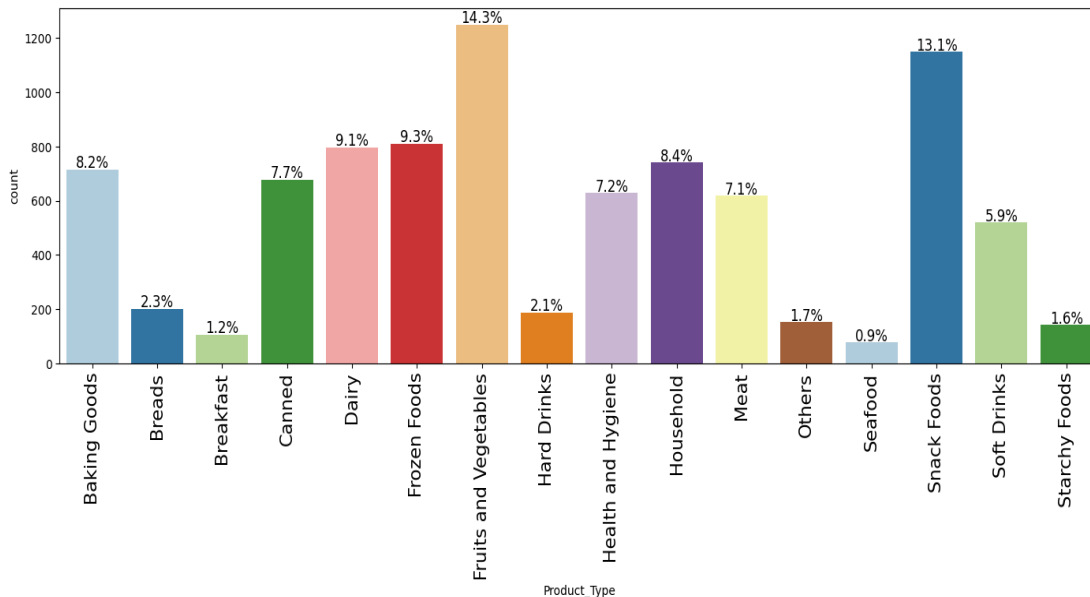
Univariate Analysis: Product Sugar Content



- Low Sugar products dominate the dataset, accounting for 55.7% of all products.
- Regular sugar products represent 25.7% of the product mix, indicating strong presence of traditional offerings.
- No Sugar products account for 17.3%, reflecting moderate demand for sugar-free alternatives.
- The “reg” category represents only 1.2%, suggesting either a very limited segment or possible data inconsistency/category duplication.

EDA Results

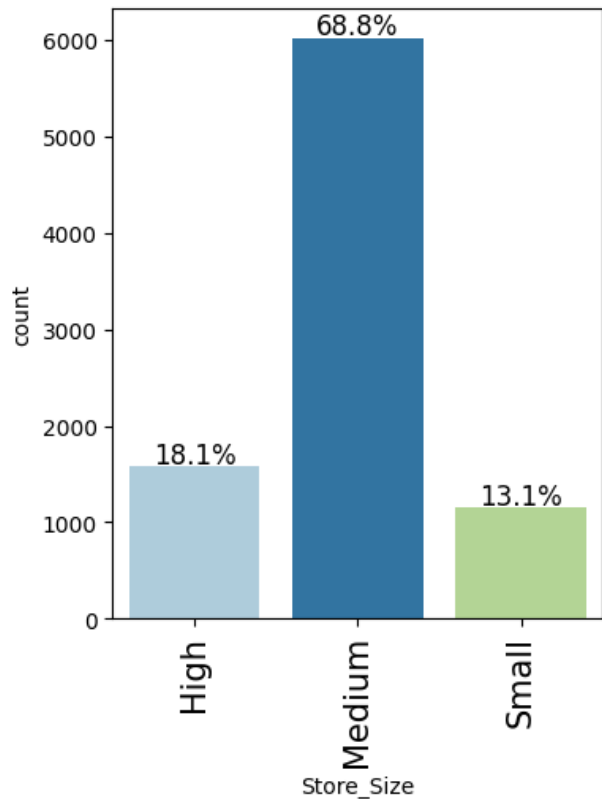
Univariate Analysis: Product Type



- Fruits and Vegetables is the largest product category, representing 14.3% of all products.
- Snack Foods account for 13.1%, making it another major category in the product portfolio.
- Frozen Foods (9.3%) and Dairy (9.1%) also contribute significantly to the assortment mix.
- Seafood (0.9%) and Breakfast items (1.2%) represent the smallest product categories.
- Product distribution is uneven across categories, indicating stronger focus on high-demand food segments.

EDA Results

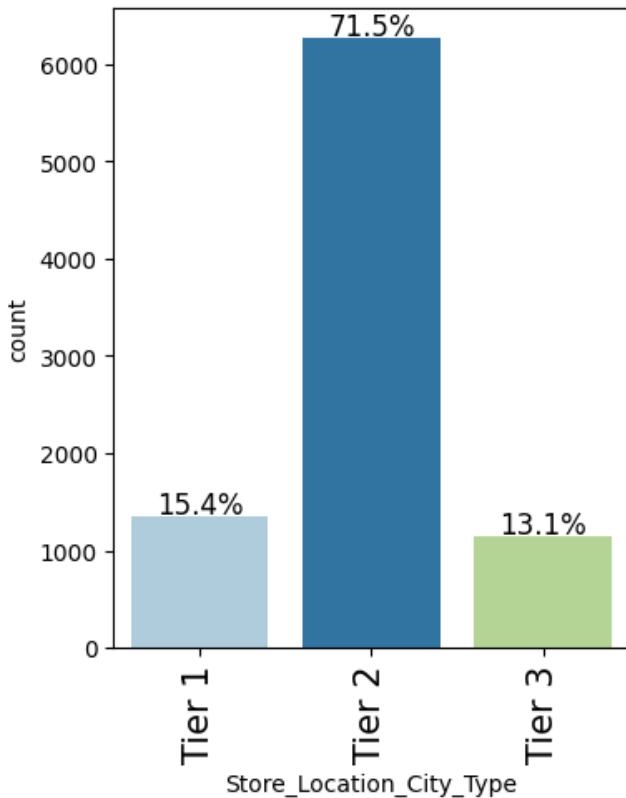
Univariate Analysis: Store Size



- Medium-sized stores dominate the dataset, representing 68.8% of all stores.
- High-sized stores account for 18.1%, while small stores represent only 13.1%.
- The store distribution is heavily concentrated toward medium-sized formats.
- The company appears to prioritize medium-sized stores as its primary retail strategy.

EDA Results

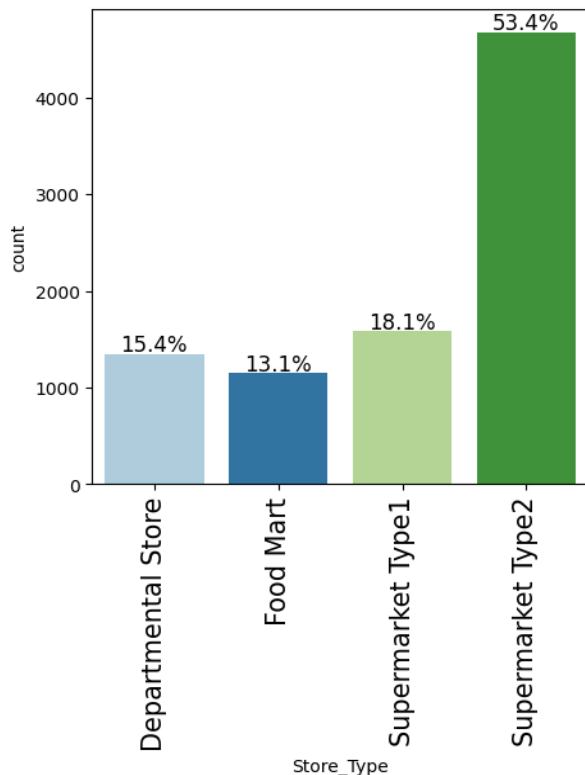
Univariate Analysis: Store Location City Type



- Tier 2 cities dominate the dataset, accounting for 71.5% of all store locations.
- Tier 1 stores represent 15.4%, while Tier 3 stores account for 13.1%.
- The company's retail presence is heavily concentrated in Tier 2 cities.
- This distribution suggests a strategic focus on mid-sized urban markets with strong consumer demand potential.

EDA Results

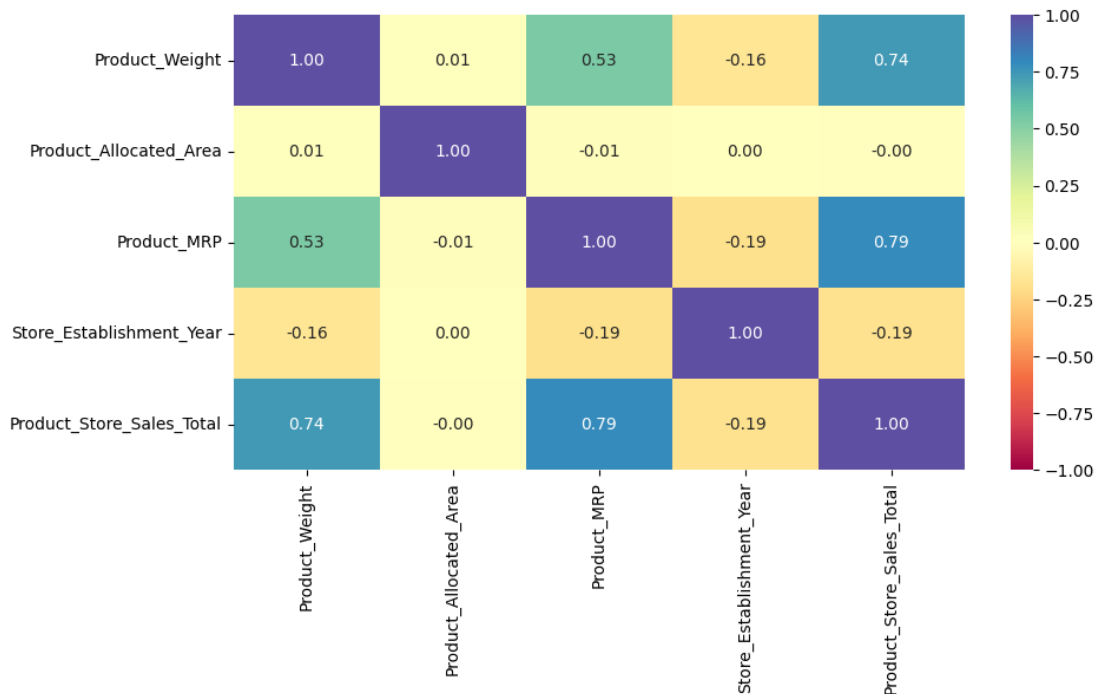
Univariate Analysis: Store Type



- Supermarket Type2 dominates the dataset, representing 53.4% of all stores.
- Supermarket Type1 accounts for 18.1%, while Department Stores represent 15.4%.
- Food Mart has the smallest share at 13.1% of total stores.
- The company's retail network is primarily concentrated in the Supermarket Type2 format, indicating it is the core operating model.

EDA Results

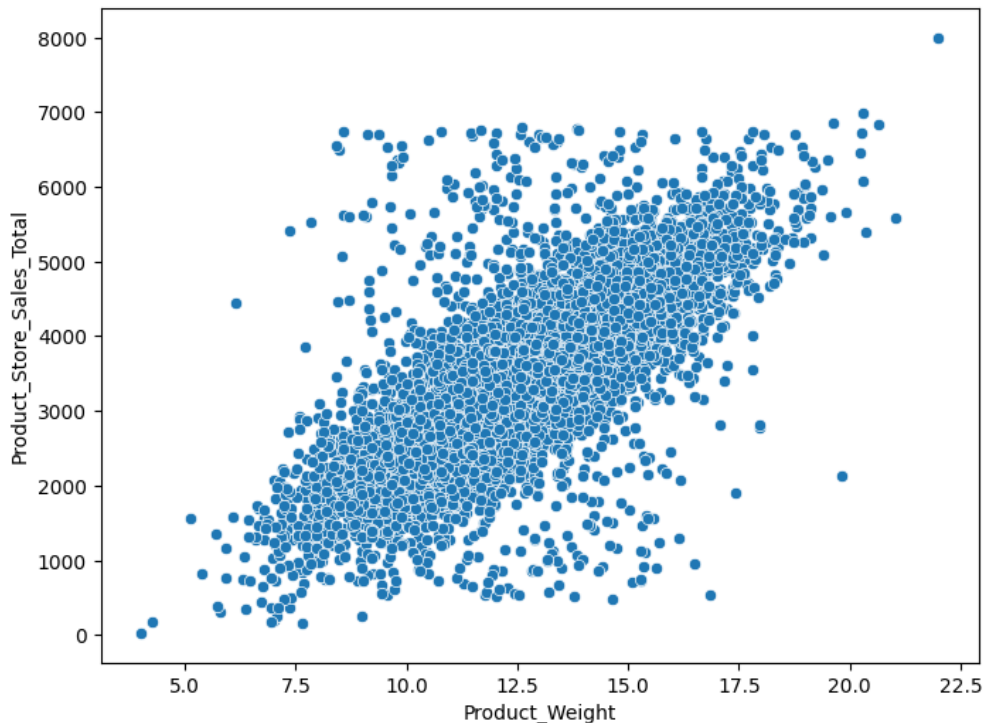
Bivariate Analysis: Correlation Matrix



- Product MRP has the strongest positive correlation with Product Store Sales Total (0.79), indicating higher-priced products tend to generate higher sales.
- Product Weight also shows a strong positive correlation with Product Store Sales Total (0.74).
- Product Weight and Product MRP are moderately positively correlated (0.53), suggesting heavier products are often priced higher.

EDA Results

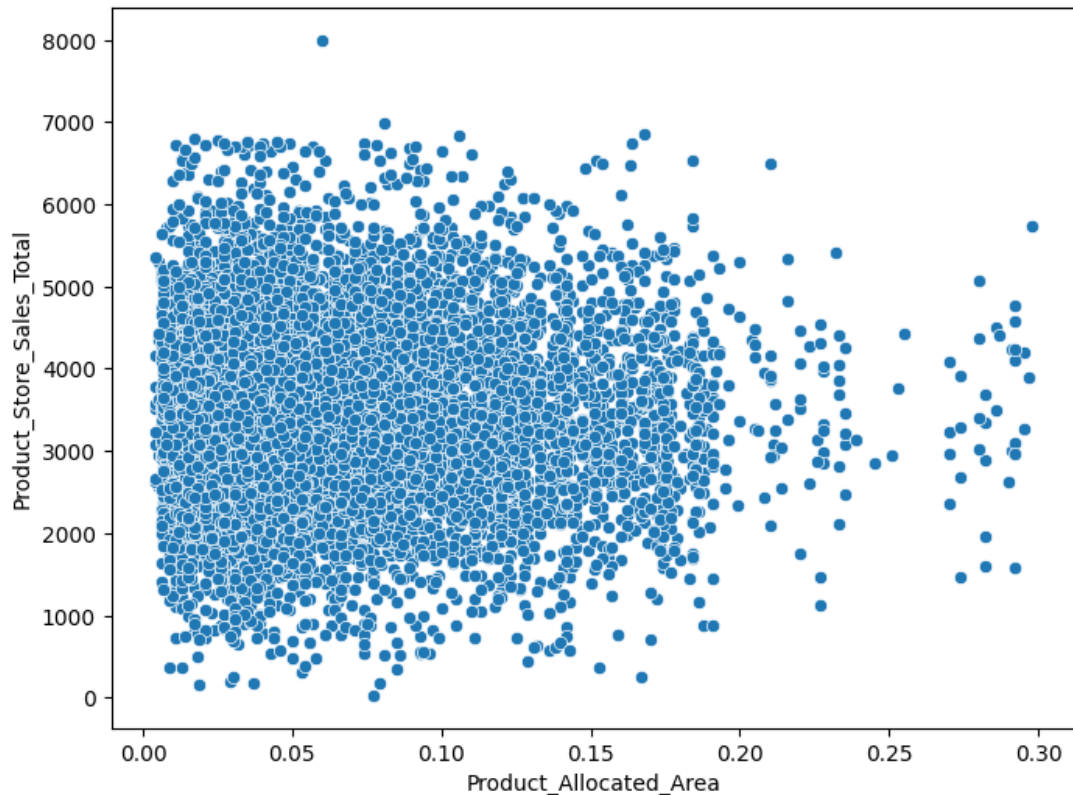
Bivariate Analysis: Product Weight vs Product Store Sales



- Product Weight shows a strong positive relationship with Product Store Sales Total.
- Higher product weights are generally associated with higher sales values.
- Most observations are concentrated between product weights of 8–18 and sales totals of 1,500–5,500.
- The scatterplot indicates Product Weight may be an important predictor of sales performance.

EDA Results

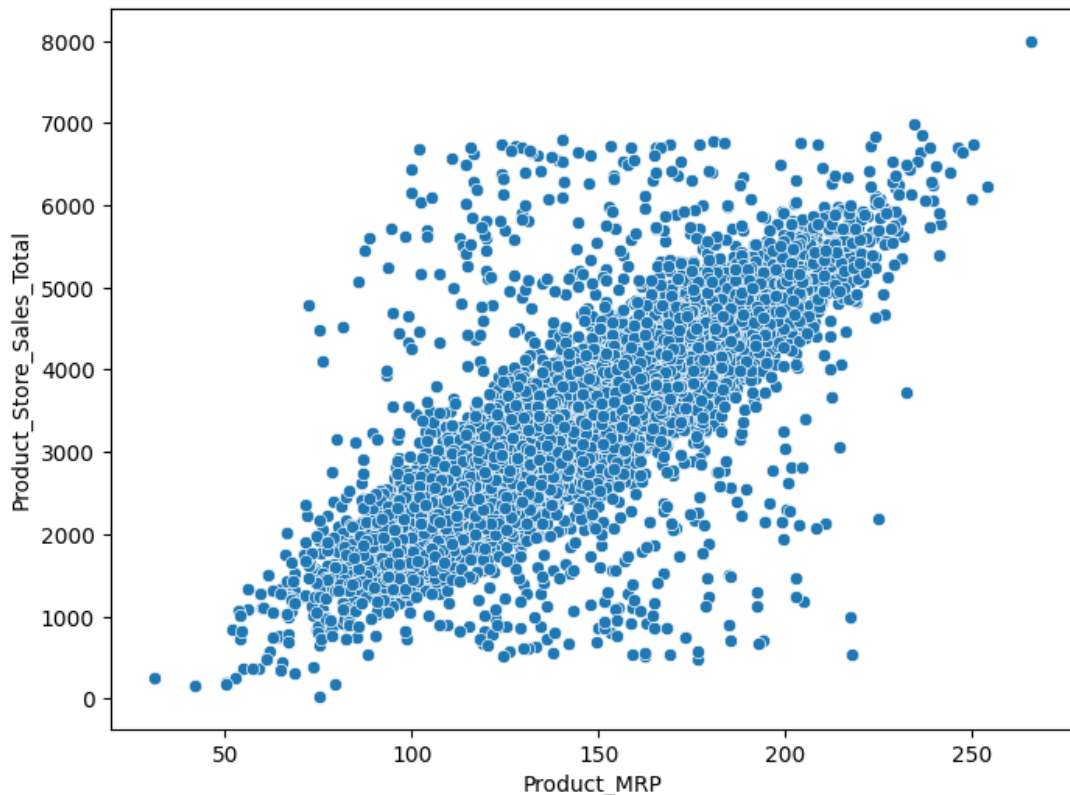
Bivariate Analysis: Product Allocated Areas vs Total Sales



- Product Allocated Area shows little to no strong linear relationship with Product Store Sales Total.
- Sales values are widely dispersed across all allocation levels, indicating weak correlation.
- Most products are concentrated within lower allocated area values below 0.15.
- A few products with large allocated areas achieve high sales, but similar sales levels are also observed at smaller allocation sizes.

EDA Results

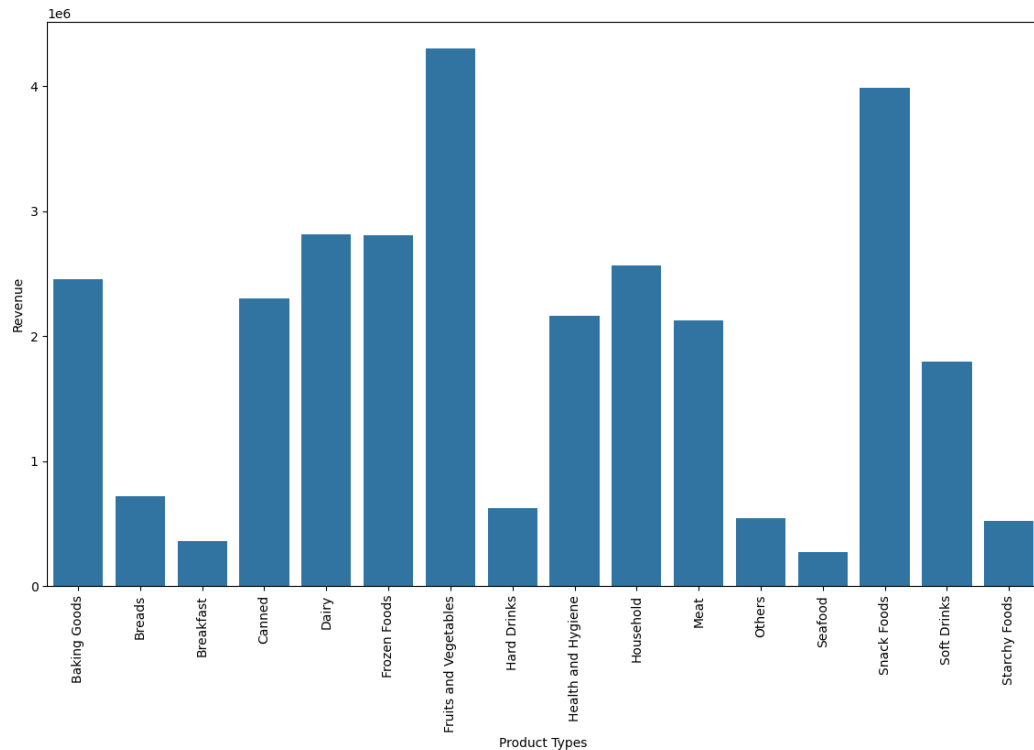
Bivariate Analysis: Product MRP Area vs Total Sales



- Product MRP shows a strong positive relationship with Product Store Sales Total.
- Higher-priced products generally generate higher sales values.
- Most observations are concentrated between MRP values of 100–220 and sales totals of 1,500–6,000.
- The scatterplot displays a clear upward trend, indicating Product MRP is a strong predictor of sales performance.

EDA Results

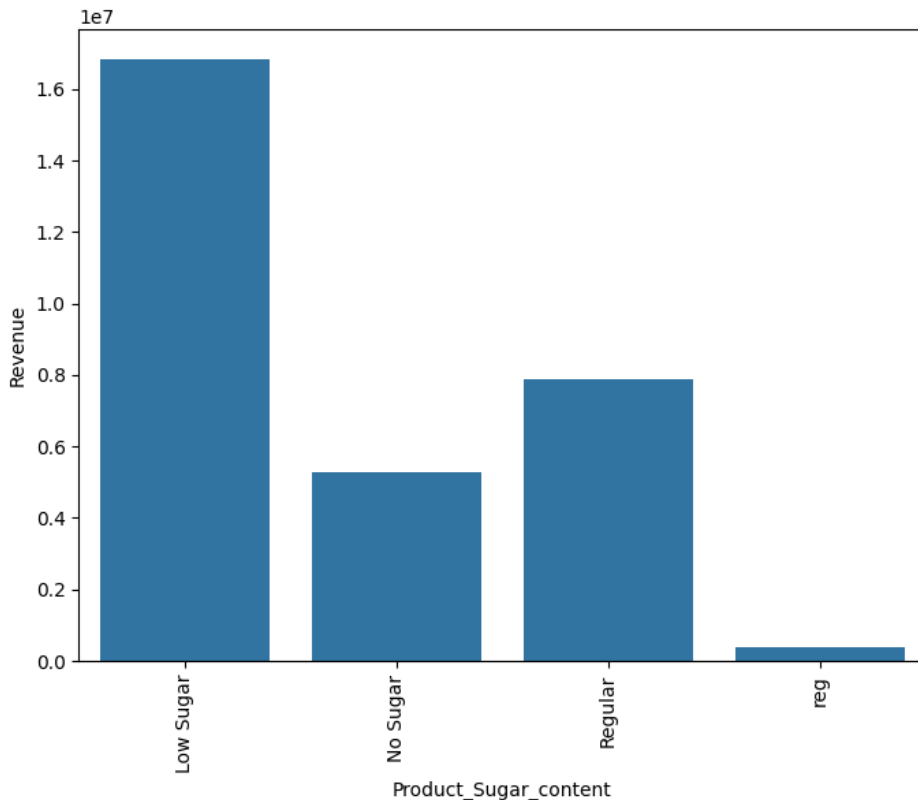
Bivariate Analysis: Revenue by Product Type



- Fruits and Vegetables generate the highest total revenue among all product categories.
- Snack Foods is the second-largest revenue contributor, indicating strong customer demand.
- Dairy and Frozen Foods also contribute significantly to overall revenue performance.
- Seafood, Breakfast, and Starchy Foods generate the lowest revenues across categories.
- Revenue distribution is concentrated among a few major product categories, suggesting key dependence on high-performing food segments.

EDA Results

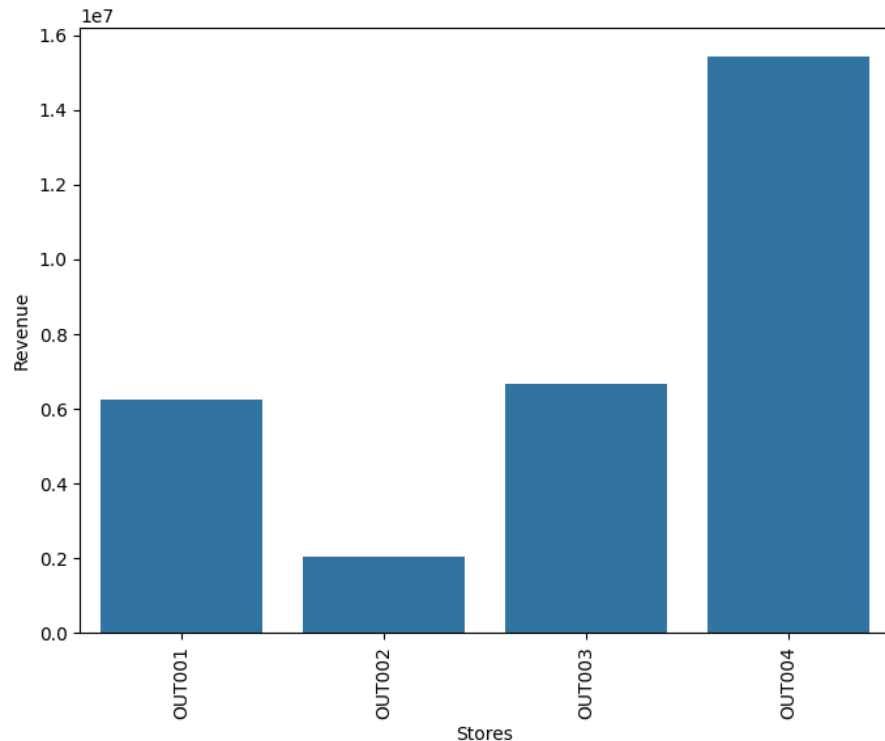
Bivariate Analysis: Revenue by Product Sugar Content



- Low Sugar products generate the highest total revenue by a significant margin.
- Regular sugar products are the second-largest revenue contributors.
- No Sugar products contribute moderate revenue compared to Low Sugar products.

EDA Results

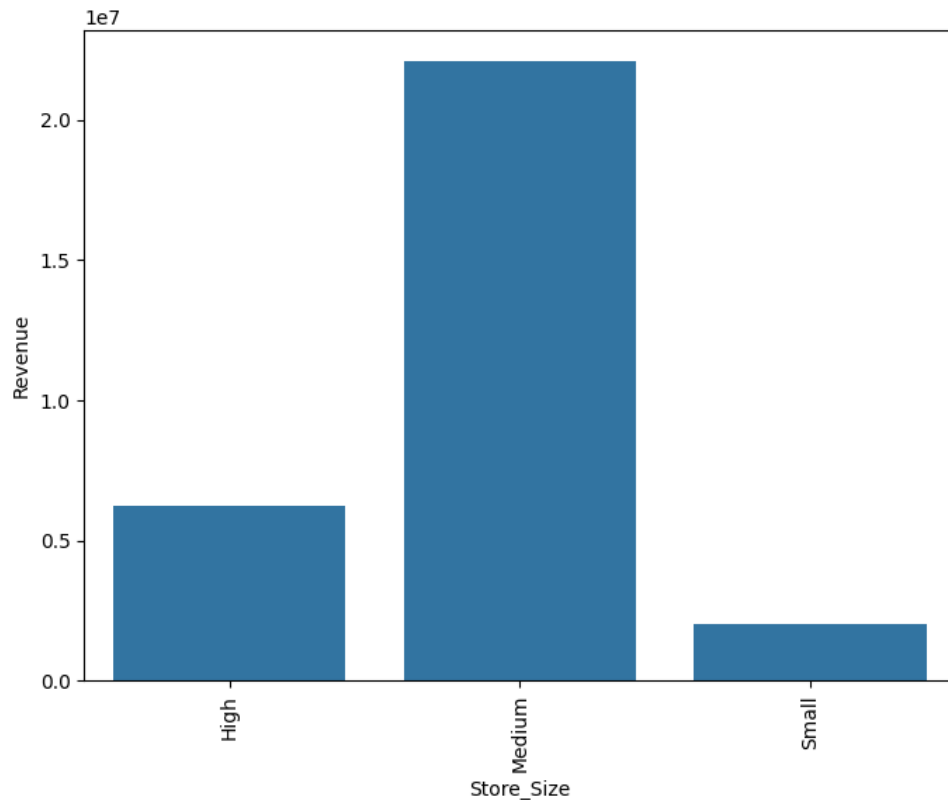
Bivariate Analysis: Revenue by Store



- OUT004 generates the highest revenue among all stores by a substantial margin.
- OUT003 and OUT001 produce moderate revenue levels, with relatively similar performance.
- OUT002 generates the lowest revenue across the four stores. Revenue distribution is uneven across store locations, indicating significant differences in store performance.

EDA Results

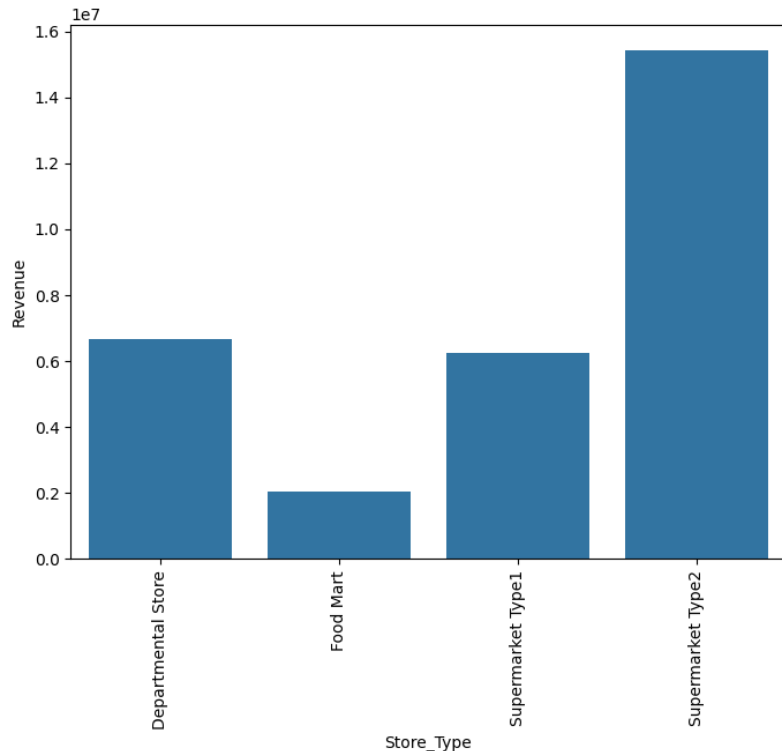
Bivariate Analysis: Revenue by Product Store Size



- Medium-sized stores generate the highest total revenue by a significant margin.
- High-sized stores contribute moderate revenue compared to medium stores.
- Small stores generate the lowest overall revenue.
- Revenue distribution aligns closely with the higher concentration of medium-sized stores in the dataset.

EDA Results

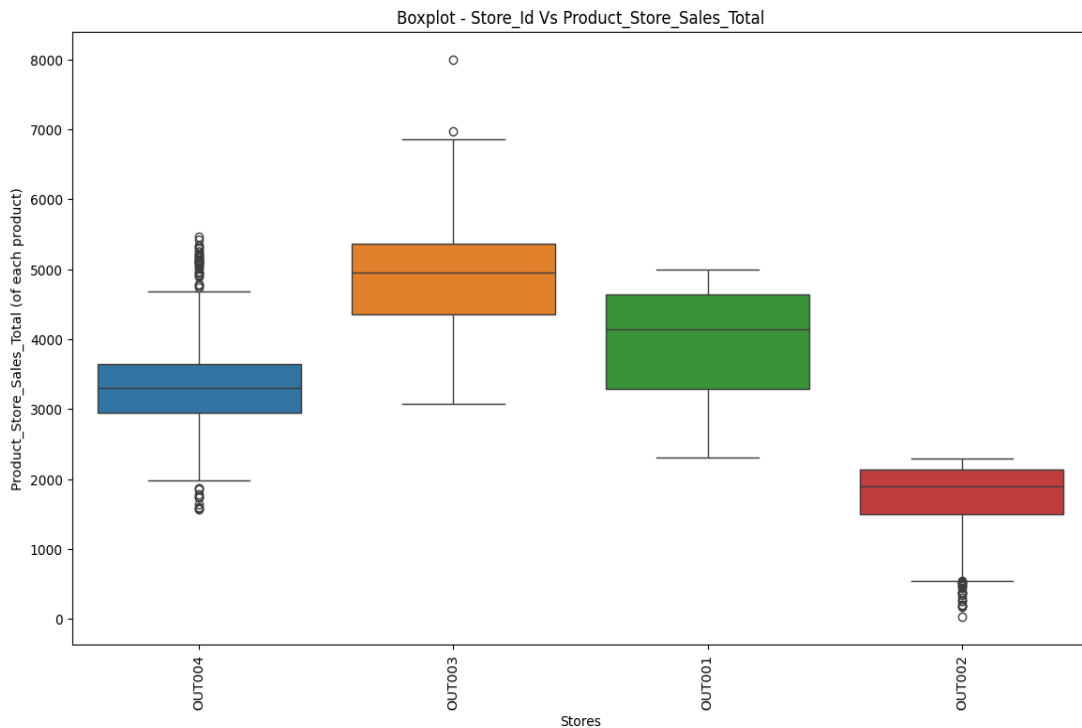
Bivariate Analysis: Revenue by Store Type



- Supermarket Type2 generates the highest total revenue among all store types by a large margin.
- Department Stores and Supermarket Type1 contribute moderate and relatively similar revenue levels.
- Food Mart produces the lowest overall revenue.
- Revenue concentration is heavily skewed toward the Supermarket Type2 format.

EDA Results

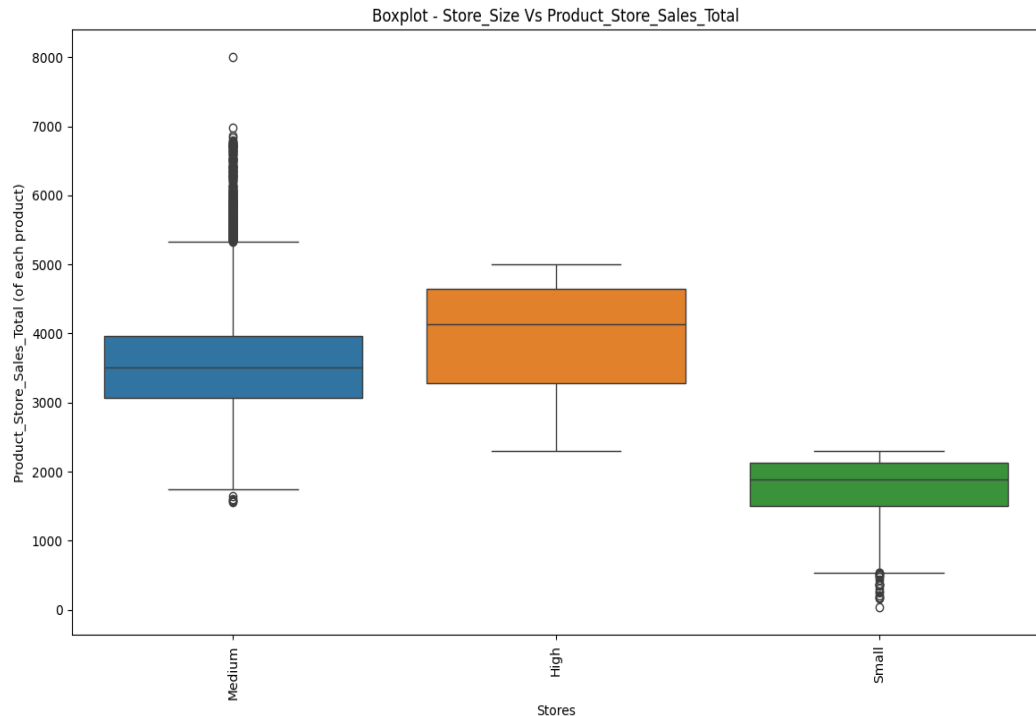
Bivariate Analysis: Store vs Product Store Sales



- OUT003 has the highest median product sales among all stores.
- OUT002 shows the lowest median sales and the narrowest sales distribution.
- OUT001 and OUT004 demonstrate moderate sales performance with relatively wider variability.
- OUT003 and OUT004 contain several high-value outliers, indicating some exceptionally strong-selling products.

EDA Results

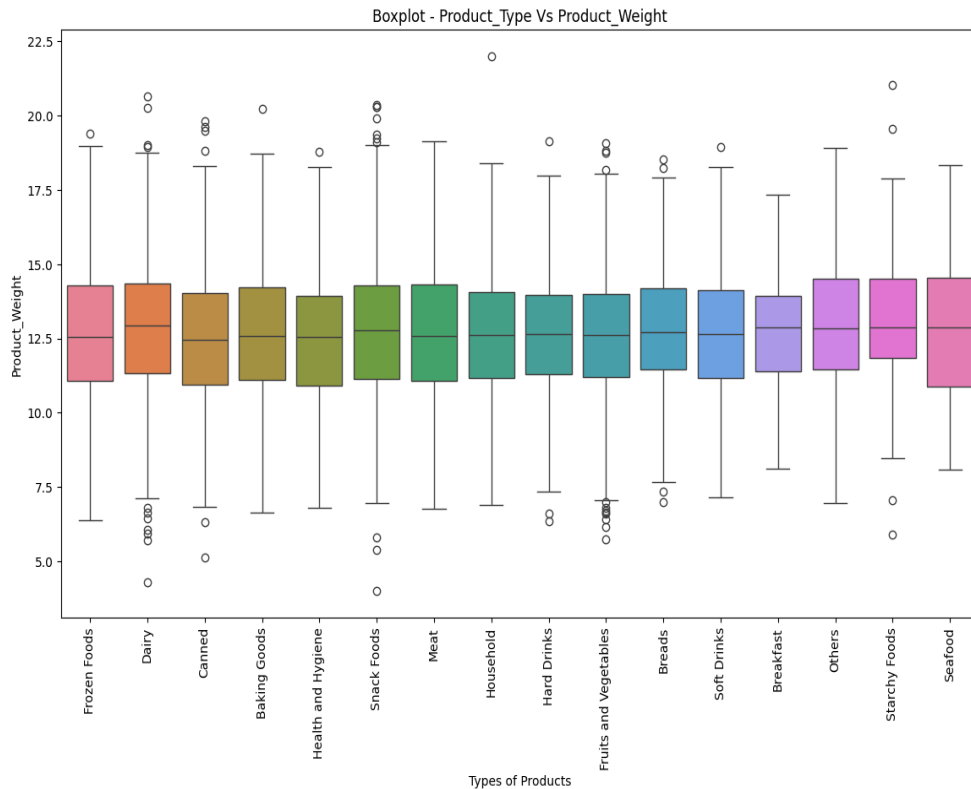
Bivariate Analysis: Store Size vs Product Store Sales



- High-sized stores have the highest median product sales among all store sizes.
- Small stores show the lowest median sales and the narrowest sales distribution.
- Medium-sized stores demonstrate moderate sales performance with a large number of high-value outliers.
- Medium stores contain the highest variability in sales, indicating inconsistent product performance across locations.

EDA Results

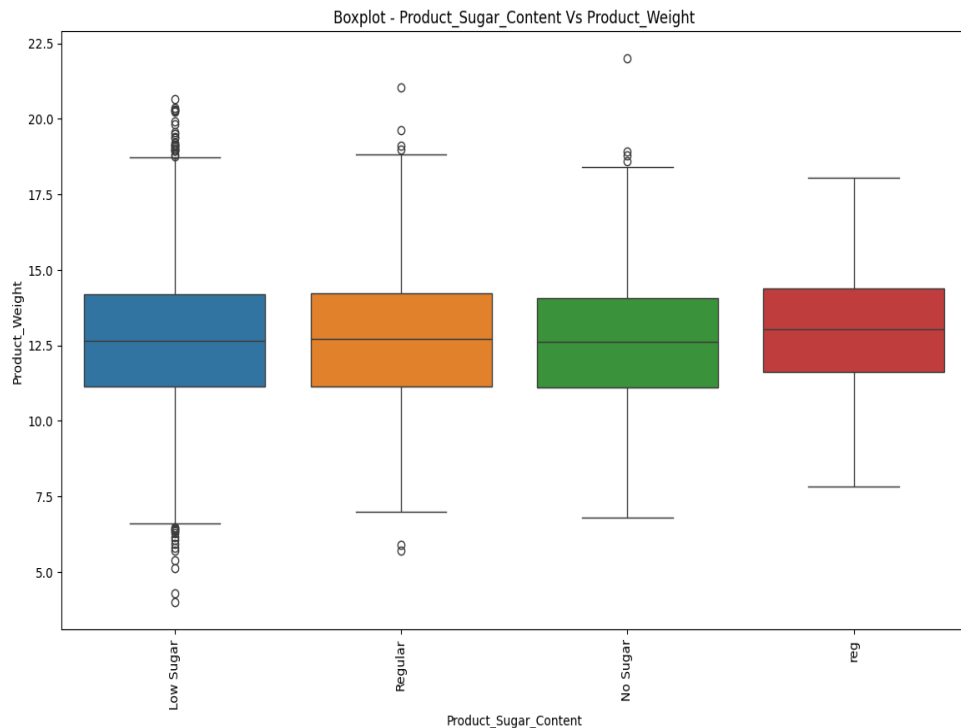
Bivariate Analysis: Product Type vs Product Weight



- Product weight distributions are relatively similar across most product categories, with medians centered around 12–13 units.
- Seafood, Starchy Foods, and Others show slightly higher median product weights compared to other categories.
- Dairy and Fruits & Vegetables exhibit several low-weight outliers.
- Most categories contain outliers on both the lower and higher ends, indicating variability in packaging sizes within categories.

EDA Results

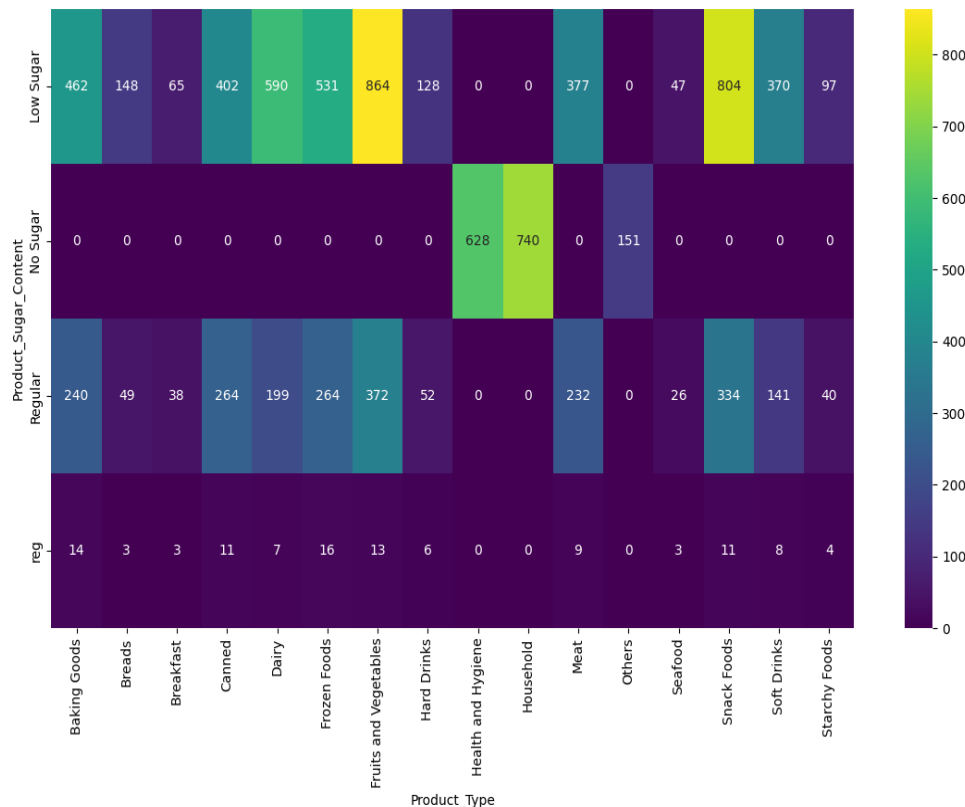
Bivariate Analysis: Sugar Content vs Product Weight



- Product weight distributions are very similar across all sugar content categories.
- The “reg” category shows a slightly higher median product weight compared to the other groups.
- Low Sugar products display the widest range of weight outliers on both the lower and higher ends.
- No Sugar and Regular categories exhibit relatively consistent weight distributions with moderate variability.

EDA Results

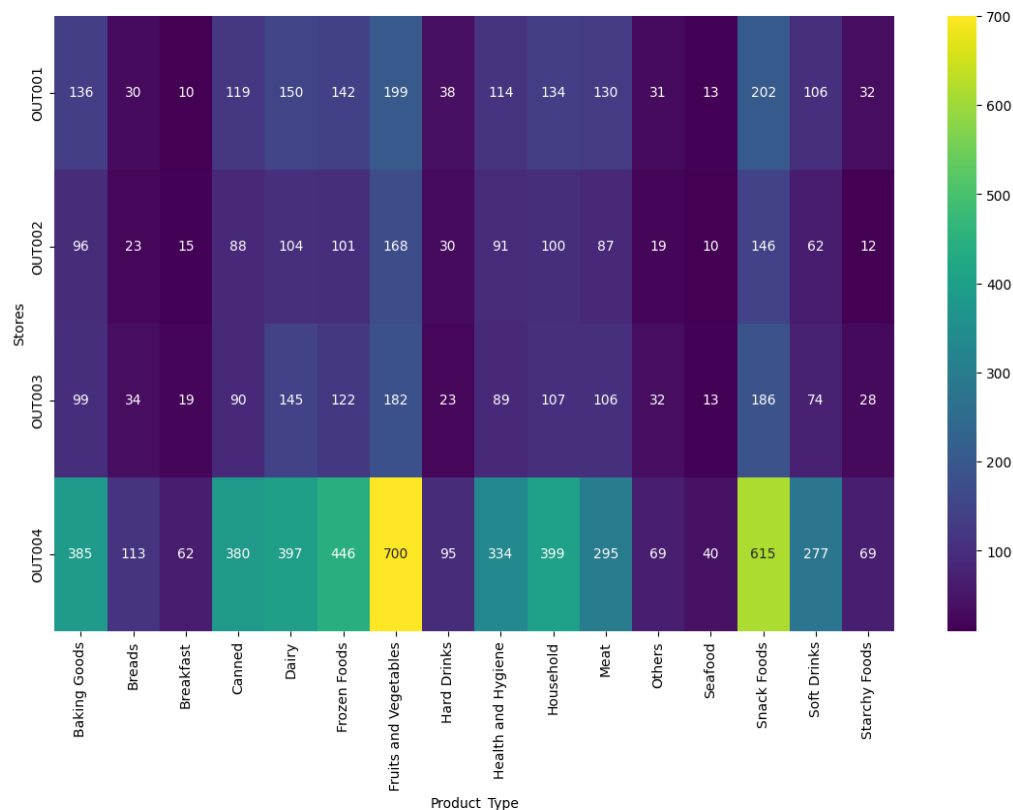
Bivariate Analysis: Sugar Content vs Product Type



- Low Sugar products dominate most product categories, particularly Fruits & Vegetables and Snack Foods.
- No Sugar products are heavily concentrated in Health & Hygiene and Household categories.
- Regular sugar products are present across many food-related categories but at lower counts than Low Sugar products.

EDA Results

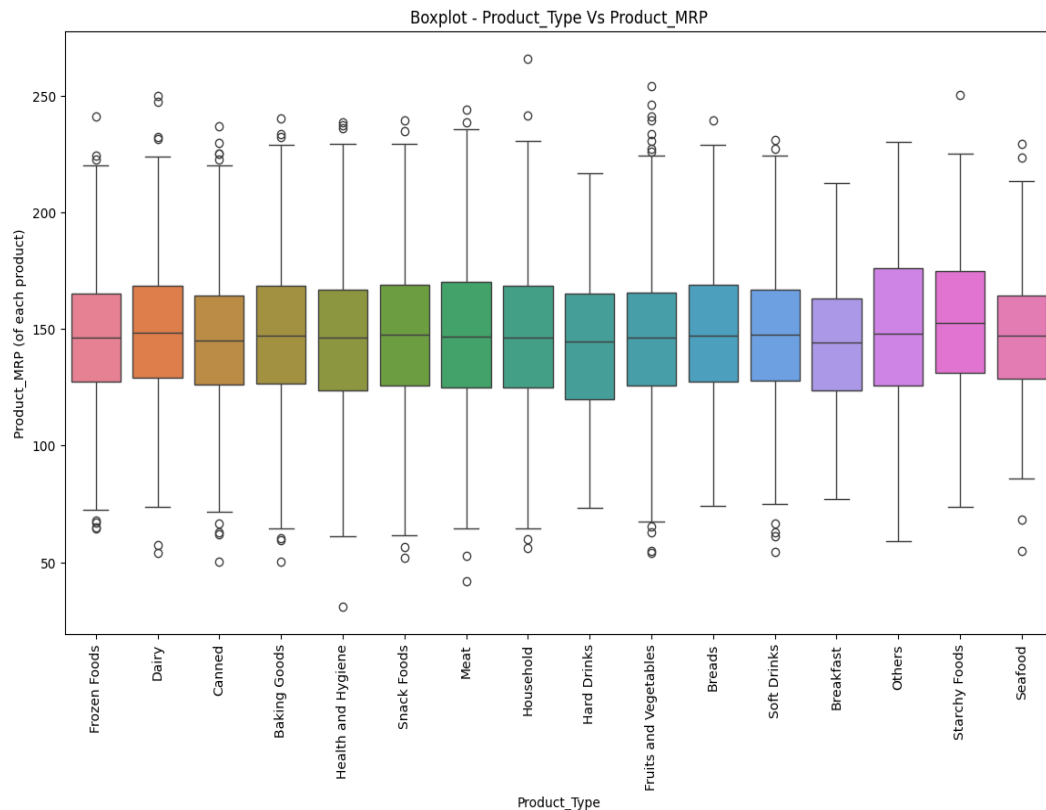
Bivariate Analysis: Product Type vs Stores



- OUT004 has the highest product counts across nearly all product categories.
- Fruits & Vegetables and Snack Foods are the most represented categories across all stores.
- OUT002 consistently shows the lowest product counts across most product types.
- Product distribution patterns are relatively similar across stores, but inventory volume differs significantly.

EDA Results

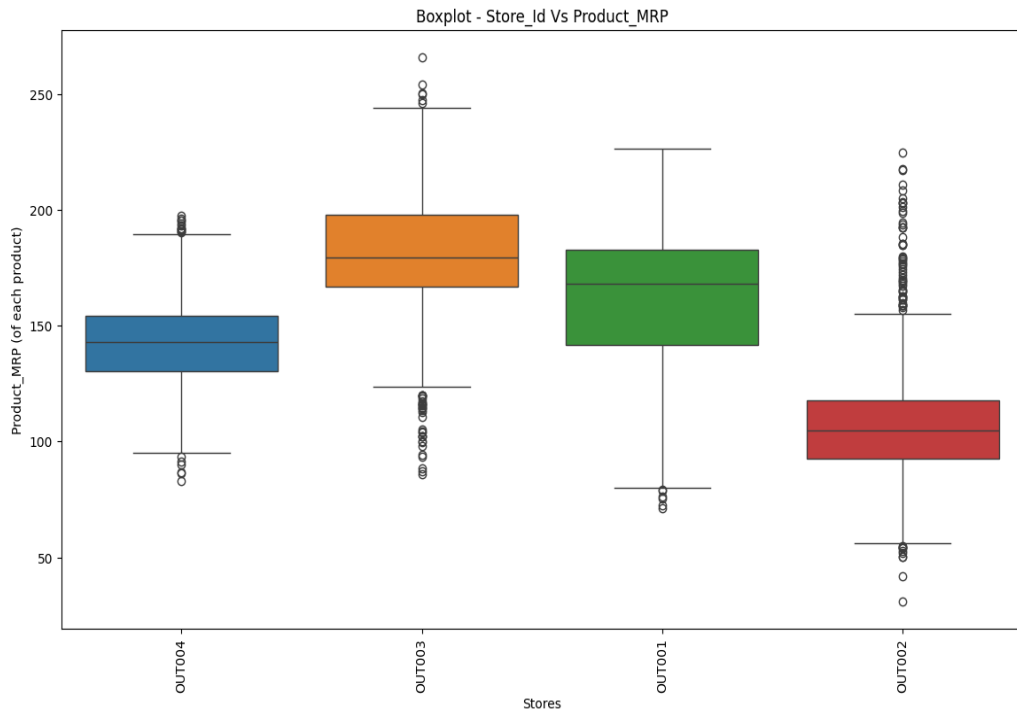
Bivariate Analysis: Product Type vs Product MRP



- Product MRP distributions are relatively similar across most product categories, with median prices generally centered around 140–150.
- Starchy Foods and Others show slightly higher median MRPs compared to other categories.
- Breakfast and Hard Drinks exhibit slightly lower median price distributions.
- Several categories contain high-price outliers above 220, indicating the presence of premium-priced products.

EDA Results

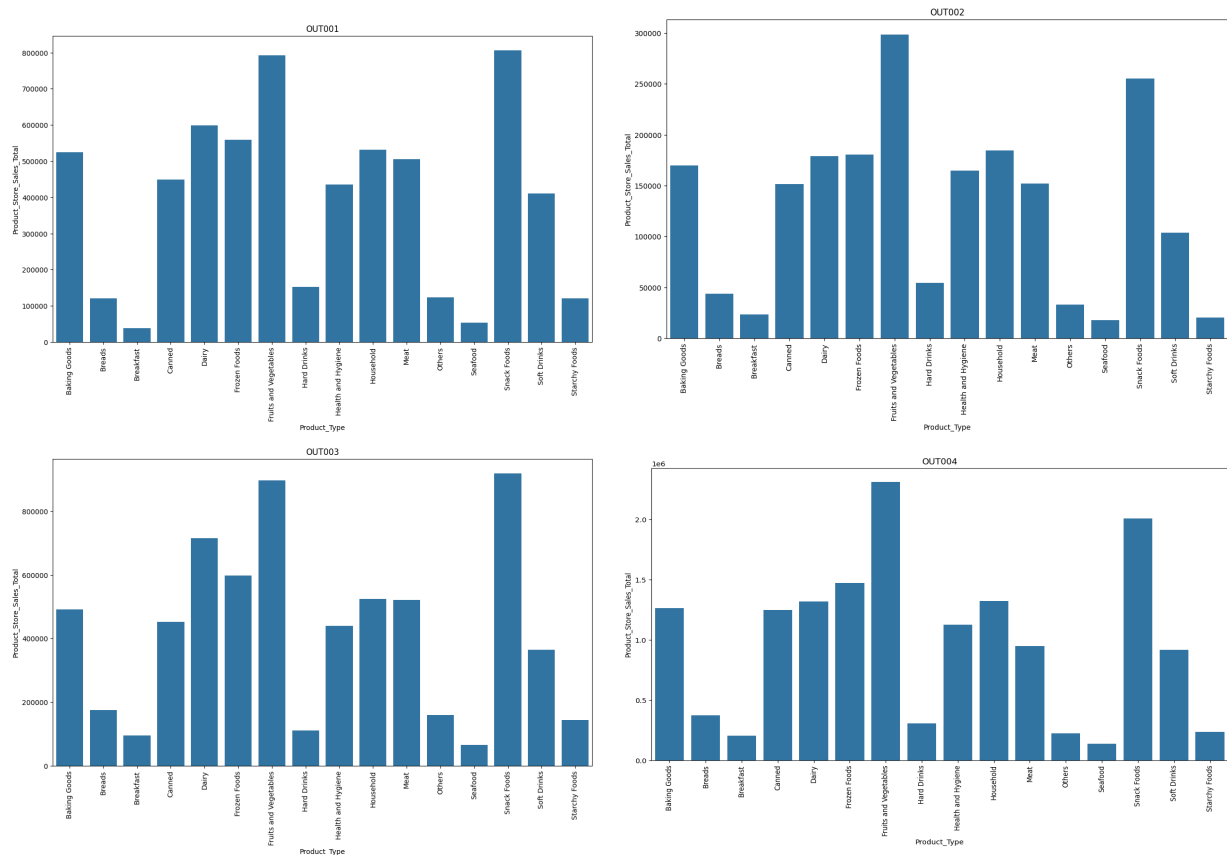
Bivariate Analysis: Boxplot Store ID vs Product MRP



- OUT003 has the highest median product MRP among all stores.
- OUT002 shows the lowest median MRP and the widest concentration of lower-priced products.
- OUT001 and OUT004 maintain moderate pricing distributions compared to the other stores.
- OUT003 contains several high-price outliers above 240, indicating a stronger presence of premium-priced products.

EDA Results

Revenue by store analysis



- OUT004 consistently outperforms all other stores across nearly every product category, making it the strongest revenue-generating store in the portfolio.
- Fruits & Vegetables and Snack Foods are the top-performing categories across all four stores, indicating they are the primary drivers of overall sales performance.
- OUT002 generates substantially lower revenues compared to the other stores, while OUT001 and OUT003 demonstrate more balanced and diversified category performance.

Description

1

Column Transformation

- ✓ Drop columns 'Product_ID', 'Product_Type', 'Store_ID', 'Store_Establishment_Year'

2

Separate Features and the target column

- ✓ The dataset is separated into features (X) and target (y), where X contains all predictor variables and y contains the sales column. (Product_Store_Sales_Total) that the model will learn to predict.

3

Split the data into train and test sets

- ✓ X_train.shape (6134, 10) → Test 70%
- ✓ X_test.shape (2629, 10) → Test 30%

4

Data Pre-processing Pipeline

- ✓ A preprocessing transformer is created using make_column_transformer and OneHotEncoder to convert categorical variables into numerical columns.
- ✓ Categorical values: (Product_Sugar_Content, Store_Size, Store_Location_City_Type, Store_Type, Product_Id_char, and Product_Type_Category)

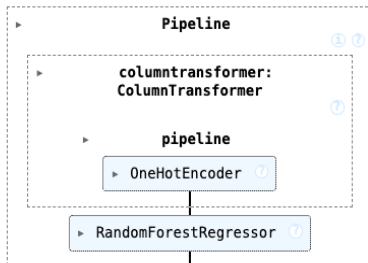
Model Building: Random Forest Model

Hyperparameter Tuning

N/A

Random Forest Regressor Model

```
rf_estimator = RandomForestRegressor(random_state=1)
rf_estimator = make_pipeline(preprocessor, rf_estimator)
rf_estimator.fit(X_train, y_train)
```



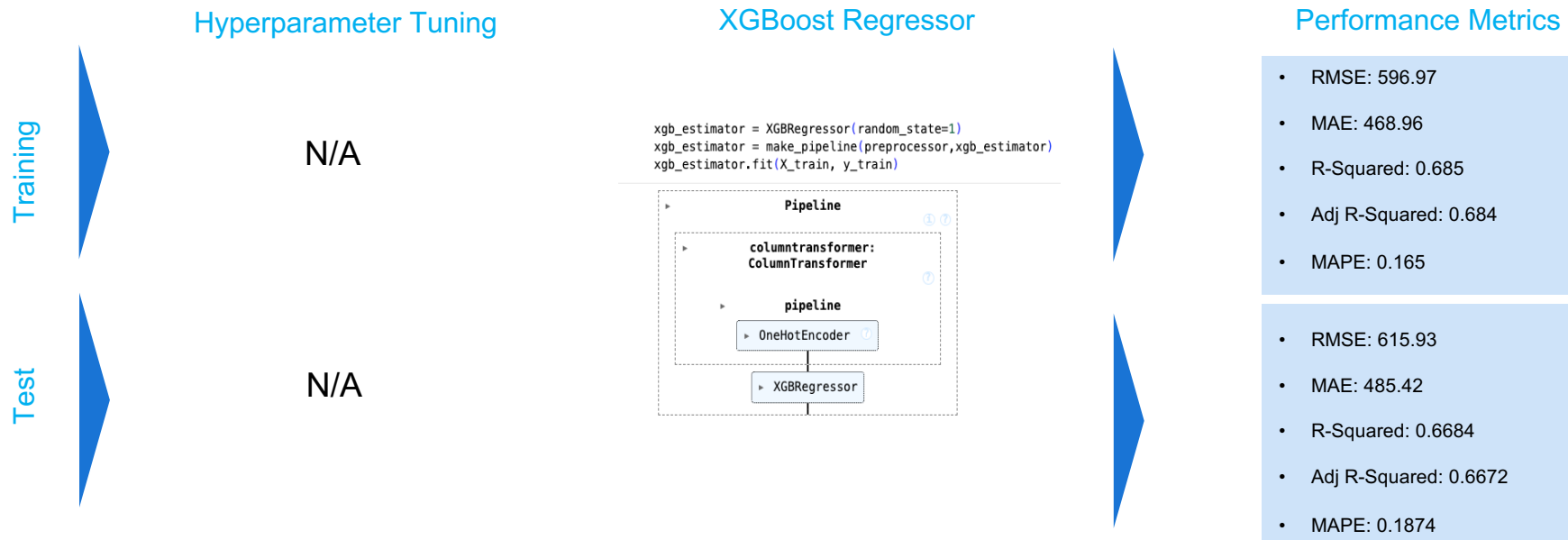
Performance Metrics

- RMSE: 596.99
- MAE: 468.87
- R-Squared: 0.685
- Adj R-Squared: 0.684
- MAPE: 0.165

- RMSE: 615.90
- MAE: 485.31
- R-Squared: 0.6685
- Adj R-Squared: 0.6672
- MAPE: 0.1873

The Random Forest Regressor model delivered stable predictive performance, achieving an R-Squared of approximately 0.67–0.69 across training and test datasets. Similar performance metrics between both datasets indicate good generalization with minimal overfitting. Overall, the model provides a reliable regression solution capable of capturing complex nonlinear relationships.

Model Building: XGBoost Regressor



The XGBoost Regressor achieved moderate predictive performance, with training and testing R-Squared values around 0.68, indicating the model explains approximately 68% of the variance in the target variable. The relatively close training and testing metrics suggest the model generalizes well with limited overfitting, demonstrating stable performance across unseen data. However, the absence of hyperparameter tuning may have limited the model's full predictive potential, as further optimization could improve RMSE, MAE, and overall accuracy.

Model Building: Tuned Random Forest Model

Training

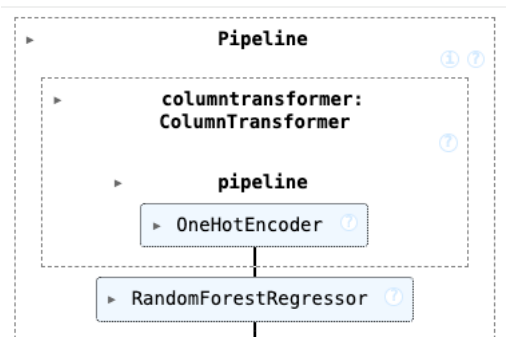
Hyperparameter Tuning

- max_depth: [5,10,15,20]
- max_features: ['sqrt', 'log2']
- n_estimators: [100, 200,30]

Test

- max_depth: [5,10,15,20]
- max_features: ['sqrt', 'log2']
- n_estimators: [100, 200,30]

Random Forest Regressor Model



Performance Metrics

- RMSE: 597.14
- MAE: 470.40
- R-Squared: 0.6848
- Adj R-Squared: 0.684
- MAPE: 0.166

- RMSE: 615.81
- MAE: 486.71
- R-Squared: 0.6686
- Adj R-Squared: 0.6673
- MAPE: 0.1876

The tuned Random Forest Regressor model achieved stable performance, with R-Squared values near 0.67–0.68 across both training and test datasets, indicating consistent predictive capability. Hyperparameter tuning improved model balance and helped minimize overfitting, as shown by the close alignment between training and testing metrics. Overall, the tuned model provides a reliable and well-generalized regression solution with moderate prediction error and strong performance stability.

Model Building: Tuned XGBoost Regressor

Training

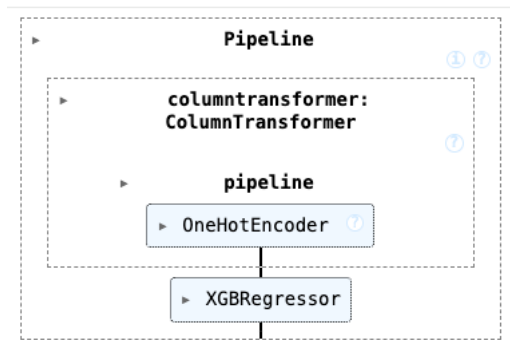
Hyperparameter Tuning

- n_estimators: [50, 100, 200]
- subsample: [0.6, 0.8, 1.0]
- gamma: [0, 0.1, 0.3, 0.5]
- colsample_bytree [0.6, 0.8, 1.0]
- colsample by level [0.6, 0.8, 1.0]

Test

- n_estimators: [50, 100, 200]
- subsample: [0.6, 0.8, 1.0]
- gamma: [0, 0.1, 0.3, 0.5]
- colsample_bytree [0.6, 0.8, 1.0]
- colsample by level [0.6, 0.8, 1.0]

XGBoost Regressor



Performance Metrics

- RMSE: 597.12
- MAE: 468.80
- R-Squared: 0.684
- Adj R-Squared: 0.684
- MAPE: 0.165

- RMSE: 615.75
- MAE: 485.01
- R-Squared: 0.6686
- Adj R-Squared: 0.6674
- MAPE: 0.1874

The tuned XGBoost Regressor achieved stable performance, with R-Squared values near 0.67–0.68 across training and test datasets. Hyperparameter tuning improved model stability while minimizing overfitting through optimized sampling and estimator settings. Overall, the model provides a reliable regression solution with balanced predictive accuracy on unseen data.

Model Performance Summary

Training performance comparison:

	Random Forest	Random Forest Tuned	XGBoost	XGBoost Tuned
RMSE	596.994959	597.143132	596.978222	597.122729
MAE	468.875850	470.405518	468.965507	468.801945
R-squared	0.685016	0.684859	0.685033	0.684881
Adj. R-squared	0.684501	0.684345	0.684519	0.684366
MAPE	0.165674	0.166015	0.165690	0.165737

Testing performance comparison:

	Random Forest	Random Forest Tuned	XGBoost	XGBoost Tuned
RMSE	615.906846	615.814107	615.933034	615.751767
MAE	485.311027	486.717211	485.429585	485.029382
R-squared	0.668510	0.668610	0.668482	0.668677
Adj. R-squared	0.667244	0.667344	0.667215	0.667411
MAPE	0.187394	0.187663	0.187421	0.187449

Model Objective

Business objective is to reduced prediction errors

- MAPE predicts sales with~81% accuracy
- RMSE shows the lowest prediction error (615.7)

XGBoost Tuned was selected because it achieved the lowest prediction error (RMSE, MAE, and MAPE) and the highest explanatory power (R^2) on the test dataset, making it the most accurate and reliable model for forecasting future sales.

Description

1

Serialization

- ✓ web app deployment folder-> `os.makedirs("backend_files", exist_ok=True)`
- ✓ `saved_model_path` -> `backend_files/superkart_xgb_tuned.joblib`
- ✓ test saved model -> `saved_model.predict(X_test)`

2

Flask Web Framework

- ✓ home page api > `@superkart_api.get('/')`
- ✓ endpoint to predict churn -> `@superkart_api.post('/v1/predict')`
- ✓ extract customer features and convert them into a DataFrame

3

Setting up a Docker Space (Hugging Face)

- ✓ `writefile backend_files` -> `requirements.txt` [pandas, numpy, joblib xgboost, flask, gunicorn, uviorn, etc..)
- ✓ docker file - > commands in Docker [WORKDIR, COPY, RUN]
- ✓ Install dependencies -> `requirements file`

4

Uploading to Docker Space (Hugging Face)

- ✓ Hugging Face import login - `HfApi` -> `access_key` & `repo_id`
- ✓ Initialize `api = HfAPI( )`
- ✓ Upload Streamlit app files -> `backend_files`, `repo_id`, `space`

5

Hugging Face Hub API

Description

- ✓ Import the login function -> `from huggingface_hub import login`
- ✓ Define access token -> `login (token = )`
- ✓ Huggingface_hub -> `create_repo`

6

Create Repository

- ✓ `create_repo` -> `"superkart-salespred"`
- ✓ `repo_type` -> `'space'`
- ✓ `space_dsk` -> `'docker'`

Uploading Files to Hugging Face

1. Define access keys and tokens:
 - access_key
 - repo_id
2. Initialize the API:
 - api=HfApi ()
3. Upload Streamlit app files:
 - frontend_files
 - repo_id
 - space
4. repo_url
 - <https://huggingface.co/spaces/ccy2k/superkart-salespred>

User Interface Intake Form

SuperKart Sales Predictor

Product Weight

12.66

– +

Product Sugar Content

Low Sugar

▼

Product Allocated Area

0.07

– +

Product MRP

150.00

– +

Store Size

Small

▼

Store Location City Type

Tier 1

▼

Store Type

Departmental Store

▼

Product ID Character

FD

▼

Store Age (Years)

15

– +

Product Type Category

Perishables

▼

Predict

Conclusions and Recommendations

Conclusions:

- EDA identified key product, store, and location factors influencing sales performance, with notable differences across product categories, store formats, and locations.
- Four regression models were evaluated: Random Forest, Tuned Random Forest, XGBoost, and Tuned XGBoost. Performance was measured using RMSE, MAE, MAPE, and R^2 .
- The Tuned XGBoost Regressor achieved the lowest prediction errors and highest R^2 score, making it the best-performing model.
- Based on its superior accuracy and ability to generalize on unseen data, the Tuned XGBoost Regressor was selected as the final model for deployment and future sales forecasting.

Recommendations:

- Deploy the Tuned XGBoost Regressor as the production model for sales forecasting.
- Regularly retrain and monitor the model using new sales data to maintain performance.
- Use forecast results to support inventory planning, demand forecasting, and operational decision-making.
- Continue enhancing the model by incorporating additional business factors such as promotions, seasonality, and market trends.



Happy Learning !

